CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION

Cambridge IGCSE™ & O Level Computer Science (0478 / 2210)

TOPIC COMPREHENSIVE QUESTION PAPER (2023 – 2026 EXAM STANDARDS)

TOPIC 1.1: Number Systems & TOPIC 1.2: Text, Sound, and Images	**Total Marks: 60 Marks • Time: 1 Hour 15 Mins**

SECTION A: NUMBER SYSTEMS (1.1) — Binary, Hexadecimal, Shifts & Two's Complement

Question 1

[Total: 6 marks]

(a) Convert the 8-bit binary number **01011010** to denary. **[1]**

Working / Answer:

(b) Convert the negative denary number **−45** into an 8-bit binary number using **two's complement representation**. Show your step-by-step working. **[2]**

Step 1 (+45 in binary):
Step 2 (Two's complement):

(c) Add the following two 8-bit binary numbers using binary addition: **[3]**

```
  0 1 0 1 1 0 1 0
+ 0 0 1 1 0 1 1 1
-----------------
```

State whether an **overflow error** has occurred and explain why.

Result:
Overflow Explanation:

Question 2

[Total: 6 marks]

(a) Convert the hexadecimal value **3B** into: **[2]**
(i) 8-bit Binary: _____ (ii) Denary: _____

(b) State **two reasons** why computer scientists use hexadecimal notation instead of binary. **[2]**

1.
2.

(c) Give **two examples** of where hexadecimal notation is used in modern computer systems. **[2]**

1.
2.

Question 3

[Total: 5 marks]

An 8-bit register contains the following binary value (Denary: 52):

0	0	1	1	0	1	0	0

(a) Perform a **logical shift of 2 places to the left** on the register contents. Write the new binary value and state the new denary value. [2]

New Denary Value: _____

(b) State the arithmetic effect of a logical left shift. Explain what happens if bits are shifted out of the most significant bit (MSB). [3]

Arithmetic effect:
Overflow/Loss of precision:

SECTION B: TEXT REPRESENTATION (1.2.1) — Character Sets & File Size

Question 4

[Total: 6 marks]

(a) Describe the difference between the **ASCII** character set and the **Unicode** character set. [3]

ASCII vs Unicode:

(b) A text document contains **2,048 characters** encoded using standard 8-bit ASCII. Calculate the exact file size of the document in **Kibibytes (KiB)**. Show your working. [3]

Working:

File Size in KiB: _____ KiB

SECTION C: SOUND REPRESENTATION (1.2.2) — Sampling, Bit Depth & Calculations

Question 5

[Total: 7 marks]

(a) Explain how analogue sound waves are converted into digital format using sound sampling. [3]

Sound Sampling Process:

(b) Define the following terms: [2]

(i) **Sample Rate:** _____

(ii) **Sample Resolution (Bit Depth):** _____

(c) Describe the effect on the **file size** and **sound quality** when the sample rate is increased. [2]

Effect on file size:
Effect on sound quality:

Question 6

[Total: 4 marks]

A mono audio track is recorded for **10 seconds** with a sample rate of **44,100 Hz** and a sample resolution of **16 bits**. Calculate the uncompressed audio file size in **Megabytes (MB)** or **Mebibytes (MiB)**. Show your full working. [4]

Formula:
Calculation:
Final File Size:

SECTION D: IMAGE REPRESENTATION (1.2.3) — Bitmaps, Pixels & Calculations

Question 7

[Total: 7 marks]

(a) Explain the terms **pixel**, **image resolution**, and **colour depth** in the context of bitmap images. [3]

Pixel:
Image Resolution:
Colour Depth:

(b) A photograph has a resolution of **1000 × 800 pixels** and uses a colour depth of **24 bits (True Colour)**. Calculate the uncompressed image file size in **Mebibytes (MiB)**. Show your working. [4]

Formula & Calculation:

File Size in MiB: _____ MiB

SECTION E: DATA COMPRESSION (1.2.4) — Lossy, Lossless & RLE

Question 8

[Total: 6 marks]

(a) Compare **lossy compression** and **lossless compression**. State one file format suitable for each type. **[4]**

Lossy Compression (Description & Example format):

Lossless Compression (Description & Example format):

(b) Explain why data compression is essential when downloading or streaming media over the internet. **[2]**

Reasons:

Question 9

[Total: 4 marks]

(a) Explain how **Run-Length Encoding (RLE)** compresses data. **[2]**

RLE Explanation:

(b) The following sequence represents pixel colour values in an uncompressed row:

```
W W W W W   B B B   W W   G G G G
```

Show how this sequence would be stored using Run-Length Encoding (RLE). **[2]**

RLE Compressed Data: